

METSAnauts — HERA Rover & Moonscape Simulation Project

Neil Rao, Arun Rebbapragada, Arnav Sangle, Advay Singi, Sanay Tyagi
Instructor: David Berry

Mission Overview

- Developed through NASA HUNCH for the Human Exploration Research Analog (HERA) program.
- Integrates an 8 ft × 8 ft lunar landscape and robotic systems to simulate lunar exploration.
- Aims to enhance realism, test robotic performance, and support crew psychological immersion.

Robotic System Overview

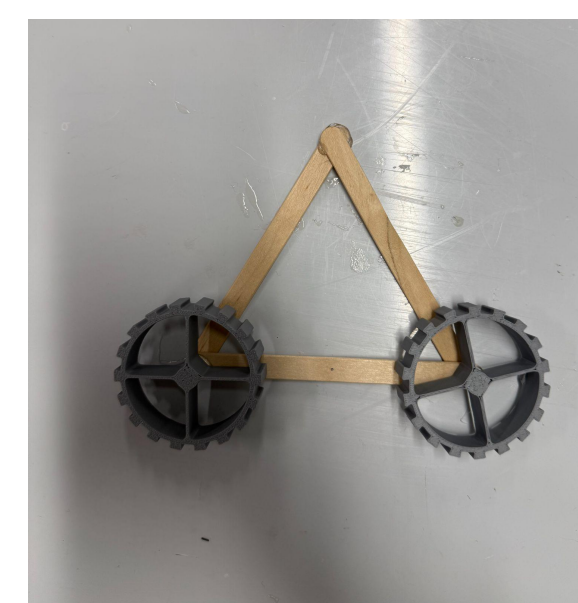
- In the initial planning phase, we planned for 4 rovers (transport rover, manipulation rover, and 2 support rovers).
- Decided to allocate our time and resources on merging all features into one rover.
- Implemented a claw to enable the rover to interact with the extraterrestrial environment.

Rover Mobility System

- 6 wheels, each powered by an independent motor (4 rear, 2 front).
- Half-tread system built for traction on granular regolith.
- Front pair lifts the chassis; rear pair maintains stability.
- Low center of gravity minimizes tilt and enhances balance.

Problem Statement

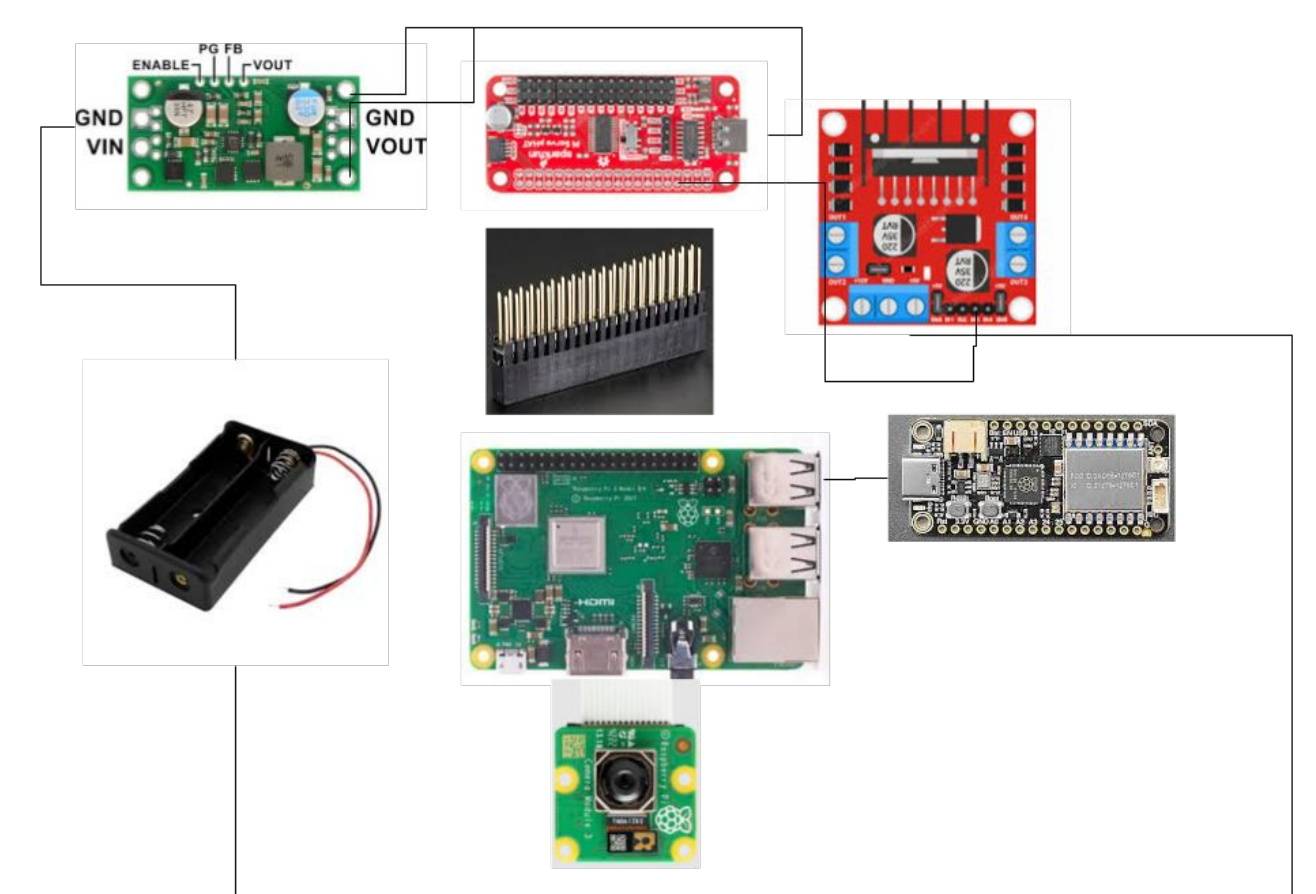
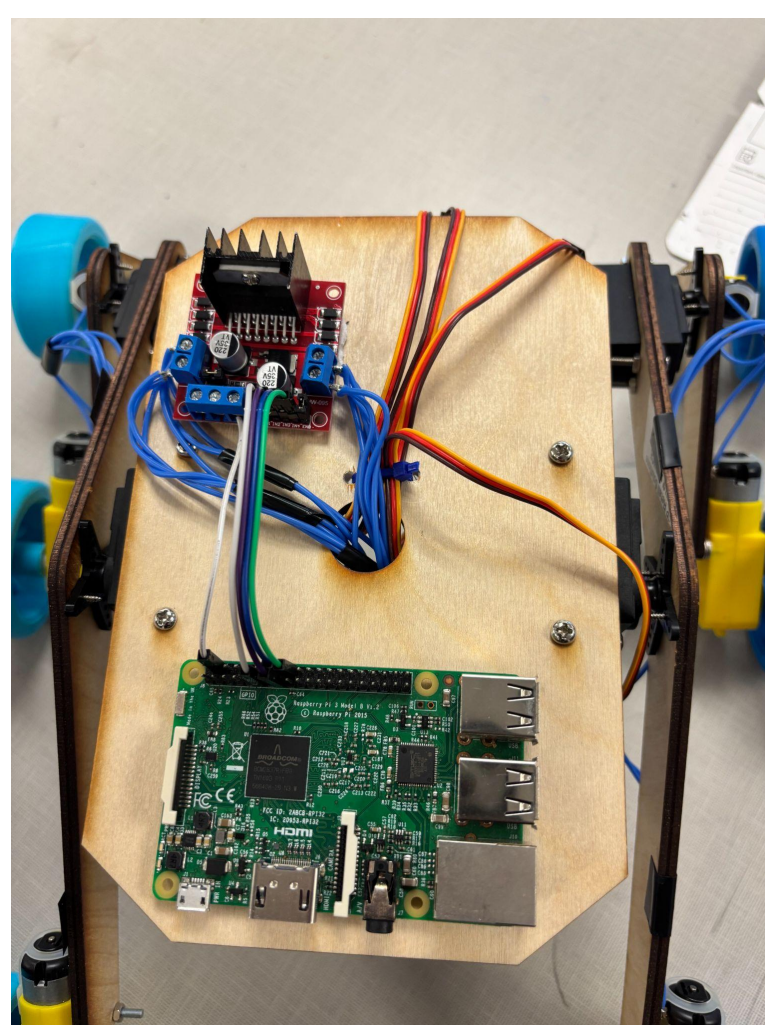
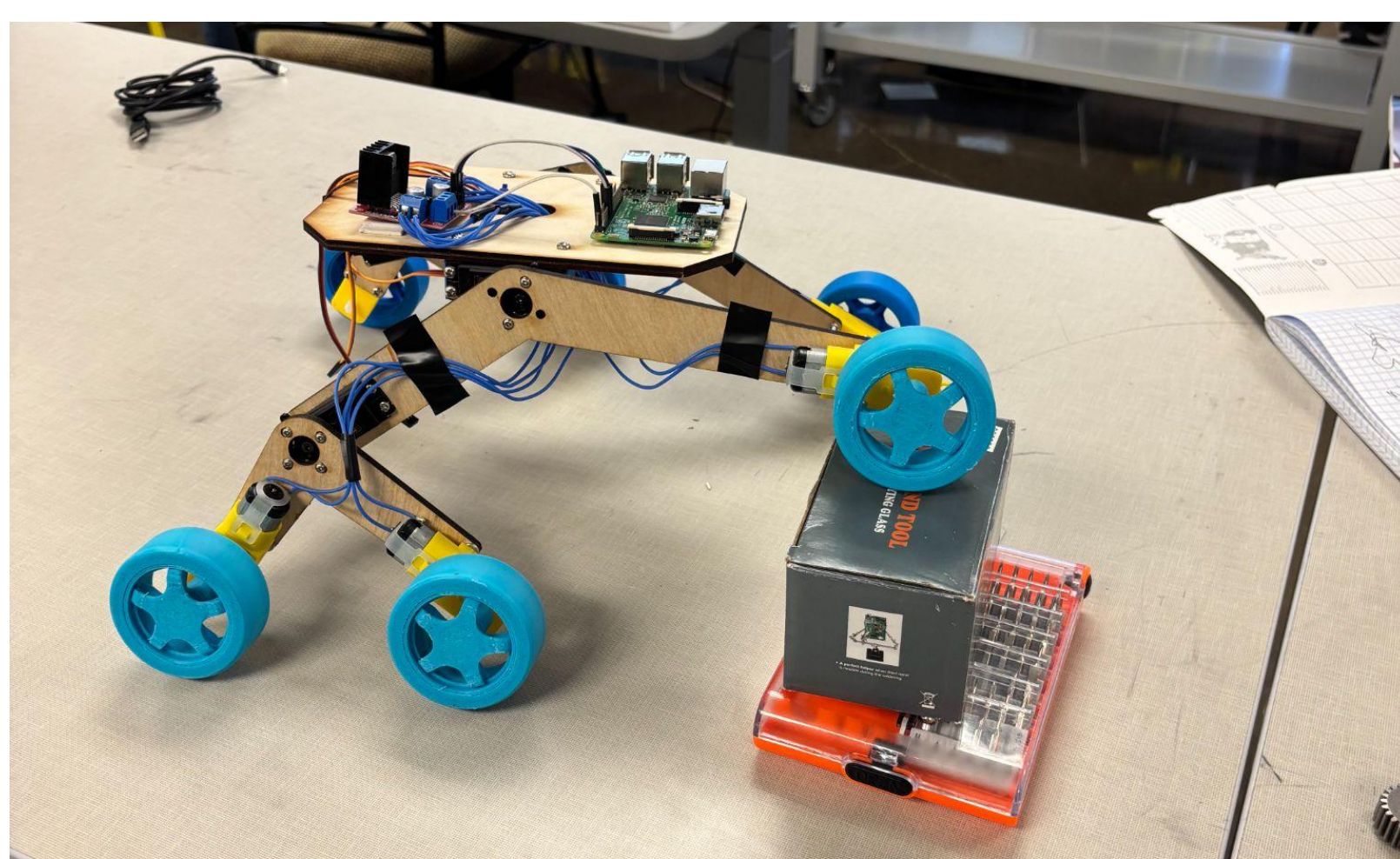
- Our challenge is to design a rover built to traverse the harsh environments of the Moon and Mars, alongside a model landscape that accurately simulates those extraterrestrial conditions.



Initial rocker bogey concept design made from popsicle sticks and 3-D printed mini-wheels

Innovations

- Solar energy utilization.
 - Uses light dependent resistors (LDRs) and orient the rover to face the sun and recharge.
- .LoRa communication
 - Useful for communication between astronaut and rover for longer distances.
- User-Friendly LoRa Communication App
 - Offers a simple, intuitive interface that lets astronauts send rover commands easily, without needing any technical background.
- Local Area Network (LAN) Connectivity
 - Provides a direct, laptop-hosted network the rover can join, ensuring reliable communication even when other links aren't available.

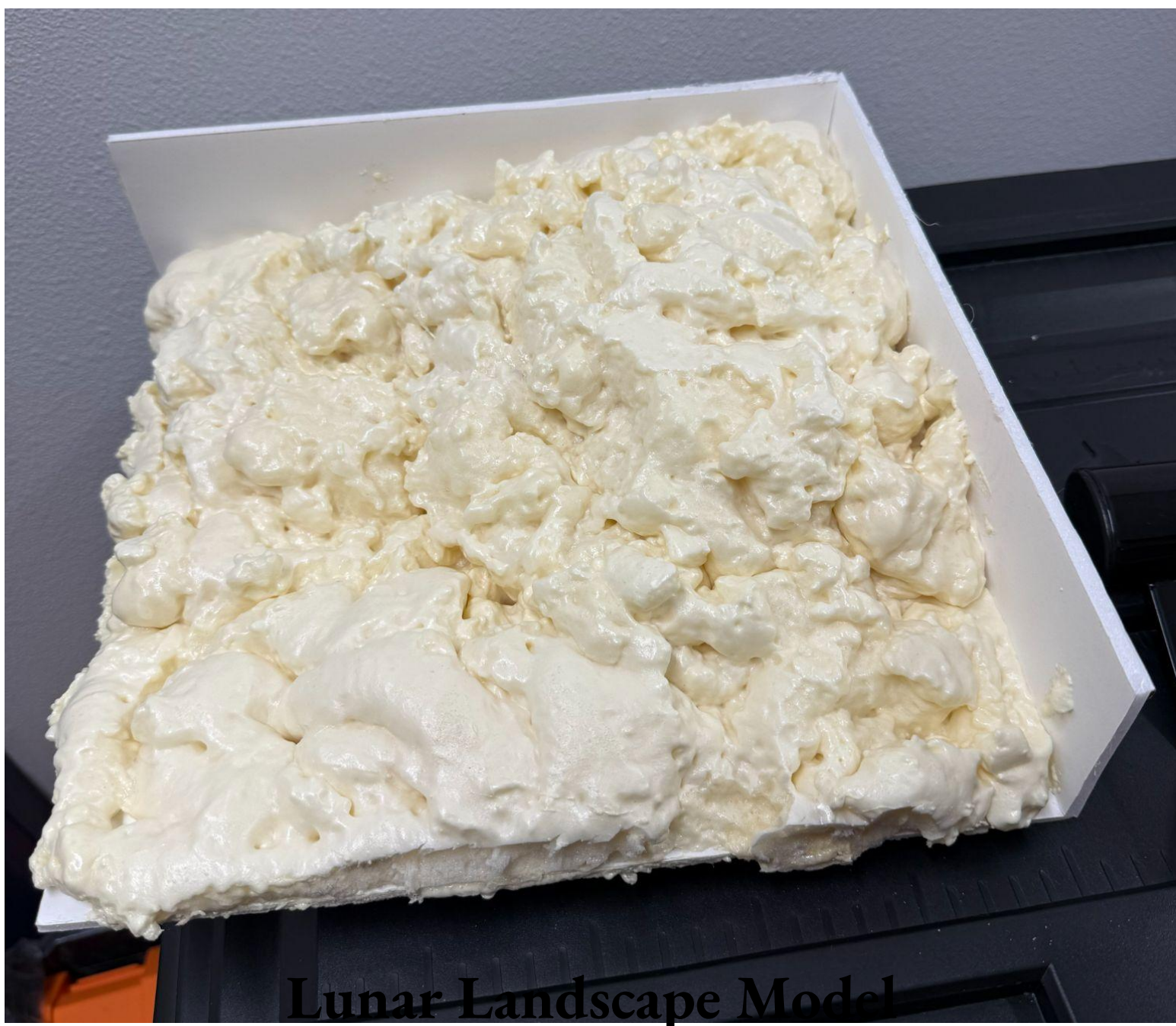


v1 Rover Design (showcasing Rocker Bogey Suspension) - Left
v1 Rover Electronic Module - Right

v1 Rover Electrical Schematic

BOTHscape Design

- Constructed with foamboard and foam gap filler terrain.
- Diatomaceous earth simulates lunar regolith.
- 2 ft x 2 ft dimensions per landscape region.
- **Moon Region**
 - Models Shackleton Crater of the Lunar South Pole.
 - Captures permanent shadow, reflective crater walls, and hydrogen deposits.
 - Provides a realistic testbed for rover navigation and power systems.
- **Mars Region**
 - Simulates Mars’ smoother, more worn-down surfaces instead of the Moon’s sharp, fresh craters.
 - Regolith resembles basaltic volcanic soil.
 - Uses finer, dustier ground shapes that match Mars better than the Moon’s chunky rocks.
 - Adds features like old channels and lava plains that don’t appear in lunar terrain.



Robotic Claw

- Designed to support mining-style tasks, allowing the rover to collect Martian regolith for testing and transport.
- Can handle loose soil, small rocks, and other surface materials without losing stability.
- Equipped with a claw-mounted camera that gives close-up visuals of targets, helping operators guide the claw precisely during collection.

Rover Communication System

- Uses WiFi for short-range communication, giving the rover a fast and stable link when operating nearby.
- Includes a simple, astronaut-friendly app that lets users send messages through the LoRa network to the rover and receive responses, even at long distances.
 - Includes a set of predefined command messages so astronauts can quickly maneuver the rover using simple, reliable inputs.
 - Each command is encoded into hexadecimal, transmitted over LoRa, decoded by the rover, and executed automatically, keeping the whole process lightweight and fast.
 - Designed with usability in mind. The interface is clean, non-technical, and straightforward, so anyone can operate the rover with no prior technical knowledge.
- Adds a redundant communication path by hosting a local area network (LAN) on a laptop, letting the rover connect directly even if WiFi or LoRa becomes unavailable.
- Supports Bluetooth for short-range video transfer from the claw-mounted camera, giving astronauts clear close-up visuals within about 20 feet.

